AE 453 ENVIRONMENTAL CONTROL AND LIVESTOCK HOUSING

ASSIGNMENT

Date released: 21 November 2018

Date due: 28 November 2018

1. A farm has 60 pigs housed in a 10 m×40 m×3 m structure with 20 m^2 of windows and 12m^2 of doors. It is 5℃ warmer inside and the heat produced per pig is 600 W. U-values in W/m^2℃ are: window - 5.8, door - 1.5, ceiling - 0.5 and wall - 0.75. What percentage of the heat production should be removed by ventilation for heat balance to occur?

2. The air temperature at noon is found to be 24 ℃ and the dew is 12 ℃. Later in the day the temperature drops to 20 ℃ without any change in humidity ratio. Find the:
    a. Relative humidity of the final state.
    b. Initial wet-bulb temperature.
    c. If the initial air above has been cooled evaporatively to saturation what would have been the final temperature.

3. A fan running at 500 rev/min on a 6 KW motor produces 12 m3/s of air at a pressure of 200 Pa. If the speed is doubled, determine
    a. the expected volume flow.
    b. Expected pressure and power.
    c. Will it be necessary to change the motor?